

Text Extraction Still Matters for End-to-End Multimodal QA Evaluation: A Comparison on ViDoRe v3

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Abstract

We study whether OCR and document parsing quality can affect end-to-end question-answering results in ViDoRe-style hybrid vision-and-text pipelines for visually rich PDFs. Although these benchmarks are often used to evaluate VLM performance in multimodal QA, their results can still reflect parser-induced bias, because the textual evidence available to the model depends directly on the parser’s extracted output. We test this effect through a controlled comparison that changes only the parsing source while holding the downstream pipeline fixed. Specifically, we parse ViDoRe v3 pages with either Nemotron Parse 1.1 or DeepSeek-OCR-2, then use the same Jina v4 text retriever, zerank-2 reranker, ColEmbed visual retriever, and Ollama-based `qwen3.5:35b` generator with a `llama3.1:8b` judge. We evaluate five public ViDoRe v3 subsets, including three English and two French subsets, using NDCG@10 for retrieval quality and pass@1 for final QA accuracy based on binary *Correct/Incorrect* judgments from a `llama3.1:8b` LLM judge. Our results show that parser choice can affect conclusions drawn from end-to-end multimodal QA evaluation, since changing only the parser reversed the end-to-end result. With Nemotron Parse 1.1, text-only QA outperformed hybrid QA (88.2 vs. 86.9 pass@1), suggesting that adding visual retrieval did not help. However, when the parser was changed to DeepSeek-OCR-2, hybrid QA outperformed text-only QA (88.7 vs. 87.6) and achieved the highest score among all English conditions, making ColEmbed appear beneficial and changing the inferred role of visual retrieval in the hybrid pipeline. French results further suggest language-dependent parser effects, since French is not included in DeepSeek-OCR-2’s reported evaluation language set, and its scores were consistently lower than Nemotron Parse 1.1 on the French datasets. Based on these findings, we argue that parser choice alone

can change retrieval and QA evaluation conclusions, and that parser-induced bias should therefore be controlled when comparing document QA systems.¹

1 Introduction

VLM performance is often evaluated in hybrid text-image QA settings, where retrieved page images are paired with parsed textual evidence. This reflects end-to-end document QA practice, where visual and textual evidence can provide complementary information (Loison et al., 2026, pp. 3–4, Table 3). Thus, VLMs are often evaluated on QA performance in settings that combine visual and textual evidence, as illustrated by the ViDoRe v3 example in Figure 1 (Faysse et al., 2025; Macé et al., 2025; Loison et al., 2026).

In this multimodal hybrid QA setting, the parser can introduce evaluation bias by determining what textual evidence is available. Unlike the visual stream, which uses page images directly, the text stream relies on OCR or document parsing before retrieval, so parser quality can affect both which textual evidence is retrieved and what textual context is ultimately provided to the QA model. However, this source of bias is usually not isolated in evaluation, and parser choice is often treated as a fixed implementation detail rather than a controlled experimental variable (Loison et al., 2026).

We therefore ask whether OCR and document parsing models can introduce bias into conclusions drawn from hybrid document QA evaluation. We answer this question through a controlled parser comparison, holding the retrieval, reranking, answer generation, and judging pipeline fixed while swapping only the parsing source between Nemotron Parse 1.1 (NVIDIA, 2024; Karmanov et al., 2025; Chumachenko et al., 2025) and DeepSeek-OCR-2 (Wei et al., 2026).

¹the corresponding codebase and result files are available in the project repository [link].

ViDoRe v3 (NVIDIA, 2024; Karmanov et al., 2025; Chumachenko et al., 2025), with DeepSeek-OCR-2 (Wei et al., 2026) while keeping the retrieval pipeline structure fixed. Unlike the original ViDoRe v3 setup, we use local open-weight models, qwen3.5:35b (Qwen Team, 2026; Yang et al., 2025) for answer generation and llama3.1:8b (Meta, 2024) for judging, and evaluate five public English and French subsets.

In our implementation, the Nemotron Parse 1.1 condition uses the built-in markdown/text extraction distributed with ViDoRe v3, while the DeepSeek-OCR-2 condition uses markdown that we newly extract from the page images.

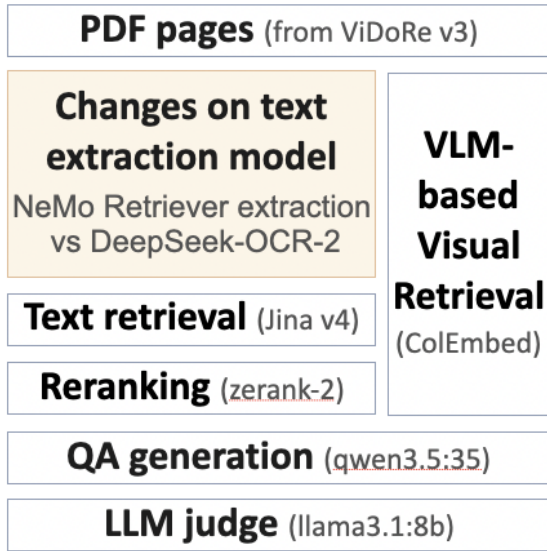


Figure 2: Overview of the controlled retrieval-and-answering pipeline. PDF pages from ViDoRe v3 flow through a text branch, a visual branch, or both depending on the experimental condition. In the text branch, we vary the extraction model by comparing Nemotron Parse 1.1 with DeepSeek-OCR-2, then retrieve with Jina v4 and optionally apply zerank-2 reranking. In the visual branch, ColEmbed retrieves page images directly from the PDFs. The QA generator receives the selected evidence stream, both streams in the hybrid setting, or no retrieved evidence in the closed-book baseline, and a separate LLM judge evaluates the generated answer.

The controlled pipeline is shown in Figure 2. Starting from five ViDoRe v3 subsets covering English and French documents, the system retrieves relevant pages through a textual stream, a visual stream, or both, and then uses the retrieved evidence to answer each query. The generated answer is then evaluated by an LLM judge, allowing us to measure the quality of the full QA pipeline.

During retrieval, the visual stream retrieves page images directly with ColEmbed (Xu et al., 2025), without an OCR or parsing step. In contrast, the textual stream first converts each page into markdown using either Nemotron Parse 1.1 (NVIDIA, 2024; Karmanov et al., 2025; Chumachenko et al., 2025) or DeepSeek-OCR-2 (Wei et al., 2026), then retrieves over the parsed text with Jina v4 (Günther et al., 2025) and optionally applies zerank-2 reranking (ZeroEntropy, 2025).

Depending on the condition, the answer generator receives retrieved markdown pages, retrieved page images, both streams, or no retrieved evidence in the closed-book baseline. It then produces a single answer, which is evaluated by a judge model as a binary *Correct/Incorrect* decision. We use qwen3.5:35b (Qwen Team, 2026; Yang et al., 2025) as the answer generator and llama3.1:8b (Meta, 2024) as the judge through Ollama (Ollama, 2026). We use different model families for generation and judging to reduce self-evaluation bias, while keeping this generator-judge pair fixed across all parser and retrieval conditions.

We then evaluate these conditions at two levels. First, we measure retrieval quality using NDCG@10 to compare the textual, visual, and reranked retrieval pipelines. Second, we evaluate end-to-end answer quality using pass@1 to measure whether the generated answer is judged correct.

4 Experiments

4.1 Datasets

We use ViDoRe v3 (Loison et al., 2026) for our experiments. The benchmark comprises 10 corpora, including seven in English and three in French, and approximately 26,000 pages, with 3,099 human-verified queries translated into six languages. All datasets are public except for two English private hold-out datasets.

In this report, we evaluate five public ViDoRe v3 subsets: three English subsets (computer_science, finance_en, and pharmaceuticals) and two French subsets (physics and finance_fr). Together, these subsets contain 1,510 queries and 10,673 document pages, spanning scientific, financial, and pharmaceutical documents with different balances of text, figures, and tables. Table 1 summarizes the evaluated subsets.

ViDoRe v3 also provides a query-difficulty anal-

Subset	Lang.	Pages	Queries	Easy queries (%)	Boolean	Compare contrast	Enumerative	Extractive	Multi hop	Numerical	Open ended
computer_science	EN	2,153	215	86.5	21	40	30	46	21	6	51
finance_en	EN	1,479	309	31.7	48	42	22	101	12	26	58
pharmaceuticals	EN	3,047	364	57.1	38	43	41	68	28	16	130
physics	FR	2,025	302	86.4	25	39	21	55	22	23	117
finance_fr	FR	1,969	320	30.0	16	52	20	86	14	45	87
Total	–	10,673	1,510	–	148	216	134	356	97	116	443

Table 1: ViDoRe v3 subsets used in our evaluation. Easy query percentages follow the ViDoRe v3 easy/hard filtering definition, where a query is easy if at least one model in a six-LLM panel answers it correctly without corpus context. Query-type counts use the primary `query_type_for_generation` label.

ysis based on parametric knowledge. In the paper, a query is classified as *easy* if at least one of six LLMs answers it correctly without any corpus context, and as *hard* otherwise³ (Loison et al., 2026). This distinction motivates our closed-book baseline and our later comparison of how much retrieval helps across query types.

4.2 Baselines

We compare the Nemotron Parse 1.1 extraction used in ViDoRe v3 (Loison et al., 2026; NVIDIA, 2024; Karmanov et al., 2025; Chumachenko et al., 2025) with DeepSeek-OCR-2 (Wei et al., 2026) as an alternative parser under matched pipeline settings. Table 2 summarizes the experimental conditions. The Nemotron Parse 1.1 conditions use the built-in markdown/text extraction distributed with ViDoRe v3, whereas the DeepSeek-OCR-2 conditions use markdown that we extract ourselves from the original page images. While corresponding baseline results for Nemotron Parse 1.1 also appear in the ViDoRe v3 paper, the numbers reported here are obtained from our own local runs.

We evaluate seven evidence conditions spanning text-only, image-only, and hybrid retrieval, plus a closed-book baseline for end-to-end answer generation. Four are text-only settings, which compare Nemotron Parse 1.1 and DeepSeek-OCR-2 text retrieval with and without zerank-2 reranking. One is an image-only setting using ColEmbed. The remaining two are hybrid settings, which combine the visual stream with the reranked text stream using either Nemotron Parse 1.1 or DeepSeek-OCR-2 extraction for the textual branch.

4.3 Evaluation Methods

We evaluate the system at two levels: retrieval quality and end-to-end answer quality.

For retrieval, we use NDCG@10 to measure the ranking quality of retrieved pages. This metric evaluates whether relevant pages are placed near the top of the ranking and is well suited to comparing the text-only and image-only retrieval pipelines under a common retrieval benchmark. We do not report NDCG@10 for the hybrid setting, because hybrid retrieval is used as an answer-generation evidence setup that concatenates the top five visual pages and top five textual pages rather than producing a single unified ranked list.

For end-to-end question answering, we report pass@1. For each query, the answer generator produces a single answer conditioned on the retrieved evidence, and a judge model labels that answer as either *Correct* or *Incorrect*. Pass@1 is then computed as the proportion of queries whose generated answer is judged correct.⁴ For answer generation, we use top-5 retrieved pages per stream in all settings; accordingly, text-only and image-only conditions provide five retrieved items to the generator, while hybrid conditions provide five page images and five markdown pages for each query.

4.4 Experimental Setup and Details

Our implementation uses a local pipeline built around Ollama. For answer generation, we use `qwen3.5:35b` as the multimodal generator, and for answer judging, we use `llama3.1:8b`. The experiments reported here use English queries for the English subsets and French queries for the French subsets.

For answer generation, we use top-5 retrieved pages per stream. Text-only and image-only settings therefore provide five retrieved items to the generator, while hybrid settings provide five page images and five markdown pages for each query,

³The six models used are GPT-5-nano, GPT-5-mini, GPT-5, Qwen3-VL-30B-A3B, Gemini 2.5 Flash, and Gemini 2.5 Pro.

⁴ViDoRe v3 describes end-to-end evaluation in binary *Correct-Incorrect* terms, although the judge prompt shown in the appendix of the same paper uses a three-way *Correct-Partially Correct-Incorrect* rubric. We follow the binary formulation.

Condition	Modality	Parser	Reranker	Top- <i>k</i>
closed_book	none	–	–	–
jina_nemo	text	Nemotron Parse 1.1	–	5
jina_nemo_reranked	text	Nemotron Parse 1.1	zerank-2	5
jina_deepseek	text	DeepSeek-OCR-2	–	5
jina_deepseek_reranked	text	DeepSeek-OCR-2	zerank-2	5
colembed	image	–	–	5
hybrid_nemo	hybrid	Nemotron Parse 1.1	zerank-2 (text stream)	5+5
hybrid_deepseek	hybrid	DeepSeek-OCR-2	zerank-2 (text stream)	5+5

Table 2: Experimental conditions. Single-stream retrieval uses the top five pages; hybrid retrieval concatenates the top five visual pages and top five textual pages without deduplication.

without deduplication across streams. For reranked text conditions, we first retrieve dense candidates and then apply zerank-2 over the top 100 candidates before selecting the final top-5 pages.

To control input length, each retrieved markdown document is truncated to at most 4,000 characters before being passed to the generator. Retrieval rankings are cached in advance, and generated answers and judgments are stored per query so that interrupted runs can be resumed without recomputation.

4.5 Results and Analysis

4.5.1 Results

The results reported below were obtained with our implemented pipeline.

Table 3 reports NDCG@10 for the text-only and image-only retrieval settings across the English and French subsets. Nemotron Parse 1.1 consistently outperforms DeepSeek-OCR-2 in text retrieval, both in English and French. With zerank-2 reranking, Nemotron Parse 1.1 reaches 75.09 average NDCG@10 on the English subsets and 51.43 on the French subsets, compared with 71.02 and 47.33 for DeepSeek.

Table 4 reports pass@1 for the closed-book, text-only, image-only, and hybrid end-to-end answer-generation settings. The English results show the clearest parser-dependent reversal. With Nemotron Parse 1.1, reranked text-only QA outperforms hybrid QA (88.2 vs. 86.9), suggesting that adding visual retrieval does not improve answer quality. With DeepSeek-OCR-2, however, hybrid QA outperforms reranked text-only QA (88.7 vs. 87.6) and achieves the highest average pass@1 among the English conditions. The French subsets show a different pattern, where `jina_nemo_reranked` remains strongest at 86.5%, suggesting that the effect is not uniform across languages.

4.5.2 Analysis

From these results, we draw two main observations.

First, parser choice can change the experimental conclusion. In our results, the answer to whether visual retrieval improves hybrid QA depends on the parser used to construct the text stream. When Nemotron Parse 1.1 is used, reranked text-only QA achieves 88.2 pass@1, while hybrid QA achieves 86.9, suggesting that adding ColEmbed does not improve QA performance. In contrast, when DeepSeek-OCR-2 is used, reranked text-only QA achieves 87.6, while hybrid QA achieves 88.7, suggesting that the same ColEmbed addition improves performance. Since this conclusion changes when only the parser is changed and the downstream pipeline is held fixed, the measured effect of visual retrieval cannot be interpreted independently of the parser used in the text stream.

Second, parser effects appear to be language-dependent. The parser-dependent reversal observed on the English subsets does not appear on the French subsets, where Nemotron Parse 1.1-based conditions are consistently stronger. In particular, DeepSeek-OCR-2-based conditions remain below the corresponding Nemotron Parse 1.1 conditions on the French datasets, and the strongest French result comes from Nemotron-based reranked text-only QA. This pattern is consistent with the reported evaluation scope of the two parsers, since Nemotron Parse 1.1 reports evaluation across several languages including French, while DeepSeek-OCR-2 is reported mainly for English and Chinese. This suggests that parser-induced bias is not simply a matter of one parser being better on average, but may depend on parser-specific properties such as language coverage, training data, and optimization targets.

Based on these observations, we make two practical recommendations for hybrid QA evaluation.

Condition	English				French			Overall Avg.
	CS	Fin. EN	Phar.	EN Avg.	Phys. FR	Fin. FR	FR Avg.	
jina_nemo	65.23	50.09	58.60	57.97	43.69	34.54	39.12	50.43
jina_nemo_reranked	83.02	72.73	69.53	75.09	49.07	53.78	51.43	65.63
jina_deepseek	64.03	46.94	56.48	55.82	40.60	31.22	35.91	47.85
jina_deepseek_reranked	82.37	65.65	65.05	71.02	46.57	48.08	47.33	61.54
colembed	78.04	68.60	67.23	71.29	47.78	47.58	47.68	61.85

Table 3: NDCG@10 for the text-only and image-only retrieval settings across English and French ViDoRe v3 subsets.

Condition	English				French			Overall Avg.
	CS	Fin. EN	Phar.	EN Avg.	Phys. FR	Fin. FR	FR Avg.	
jina_nemo	91.6	79.6	82.4	84.5	87.1	73.8	80.4	82.9
jina_nemo_reranked	92.6	83.5	88.5	88.2	92.1	80.9	86.5	87.5
jina_deepseek	90.7	73.8	82.4	82.3	83.8	73.4	78.6	80.8
jina_deepseek_reranked	95.3	79.0	88.5	87.6	90.1	80.3	85.2	86.6
colembed	93.0	74.8	83.2	83.7	83.1	70.3	76.7	80.9
hybrid_nemo	93.0	79.9	87.9	86.9	86.8	77.8	82.3	85.1
hybrid_deepseek	95.3	81.6	89.3	88.7	89.7	77.2	83.5	86.6

Table 4: Pass@1 (%) for the text-only, image-only, and hybrid end-to-end answer-generation settings across English and French ViDoRe v3 subsets.

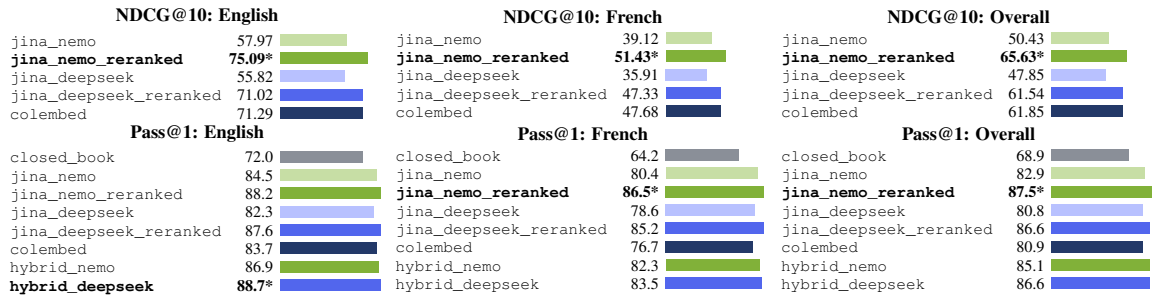


Figure 3: Summary of actual NDCG@10 and pass@1 scores by language group and overall average. Each panel is a small horizontal bar chart over the experimental conditions named in Table 2; all bar lengths use the same 0–100 score scale, and * marks the best condition within that panel.

1. Parser-induced differences should be recognized as a possible source of evaluation bias. They can change not only absolute scores, but also the conclusion drawn from a hybrid QA comparison.
2. Parser choice should be treated as an experimental variable rather than a fixed preprocessing detail. Prior ViDoRe-style evaluations vary text retrievers, rerankers, and visual retrievers, but parser choice is often kept fixed (Loison et al., 2026). Our results suggest that hybrid text-image QA systems should also be compared with parser choice controlled, or evaluated across multiple parsers.

5 Limitation

This study has several limitations because it is designed as a controlled parser comparison rather than an exhaustive benchmark of all possible hybrid document QA configurations. We intentionally fix the downstream pipeline and compare only two parsing sources in order to isolate whether parser choice can affect evaluation conclusions. As a result, our findings should be interpreted as evidence that parser-induced bias can occur under a realistic ViDoRe-style pipeline, while caution is needed before generalizing them to all settings and conditions. Specifically, this report has three main limitations:

First, our evaluation scope is limited. We evaluate only English and French subsets from the public ViDoRe v3 benchmark, so the results should not be generalized to broader multilingual settings, especially non-Latin scripts such as Chinese, which is one of the two languages included in DeepSeek-OCR-2’s reported evaluation scope.

Second, our parser comparison is limited to two systems, Nemotron Parse 1.1 and DeepSeek-OCR-2. While this is sufficient to support our claim that parser choice can affect evaluation conclusions, it does not characterize the full space of OCR and document parsing systems.

Third, we hold the other pipeline components fixed, including the retrieval model, reranker, visual retriever, answer generator, and judge. This controlled design is useful for isolating parser effects, but it also means that we cannot conclude that the same pattern would appear under every downstream configuration.

Future work can extend this comparison across more languages, datasets, parsers, retrieval models,

rerankers, answer generators, and judge models to better understand when, how, and why parser-induced bias appears. One possible next step is to test Chinese documents, since Chinese is included in DeepSeek-OCR-2’s reported evaluation scope. This was not covered in our experiments because ViDoRe v3 provides only English and French public subsets, so future work would aim to examine whether and how these effects appear beyond English and French using an appropriate Chinese document QA dataset compatible with a similar pipeline.

6 Conclusion and Future Work

We studied whether OCR and document parsing quality can bias ViDoRe-style hybrid document QA evaluation, motivated by the idea that the initial textual source extracted by a parser can affect downstream QA quality and therefore act as an experimental variable in text-image multimodal settings. Through a controlled comparison between Nemotron Parse 1.1 and DeepSeek-OCR-2, we found that parser choice can change both retrieval scores and end-to-end QA conclusions, and that these effects vary with parser properties and pipeline conditions. These findings suggest that parser-induced bias should be treated as a substantive factor in hybrid document QA evaluation, and that parser choice should be controlled or evaluated across multiple parsers when comparing text-image QA systems. At the same time, our study is limited to two parsers, five English and French ViDoRe v3 subsets, and one fixed downstream generator-judge configuration. Future work can extend this analysis to more parsers, more languages and scripts, and more downstream generator-judge configurations to better characterize when, how, and why parser-induced bias appears.

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A Contributions of Each Team Member

All team members actively participated in project discussions, reviewed intermediate results, and contributed to shaping the final direction of the project.

Alain Fidahoussen contributed to project design and led the technical implementation, including pipeline setup, retrieval and QA experiment execution, result generation, experimental result analysis, table preparation, the initial draft of the final poster, and technical content for the proposal, midway report, final presentation, and final report. **Aftab Gazali** contributed to drafting and revision of the proposal and midway report, preparation of figures and visual materials, and review of experimental materials.

Munyeong Kim contributed to project framing and led the research narrative, including proposal direction, experimental design decisions, interpretation of experimental results, development of the main argument, preparation of the final poster, writing and revision of the final report, and overall project coordination.